

	A	B	C	D	E	F	G
1	In a sample of 256 students, average marks is found to be 56 with standard deviation 12.						
2	Construct	i) 95% C.I. for Population mean.					
3		ii) 92% C.I. for Population mean.					
4		iii) confidence interval such that population mean will probably lie.					
5							
6	Solution:-Here, we have						
7	Sample size(n) =			256			
8	Sample mean ($\bar{x}$) =			56			
9	Sample SD (s) =			12			
10	S.E($\bar{x}$) =			0.75	=D9/SQRT(D7)		
11	For i) 95% C.I. for Population mean						
12	Here, C.I.(1- α) =			0.95			
13	α =			0.05	=1-D12		
14	Z_{α} =			1.95996398	=NORM.S.INV(1-D13/2)		
15	Here,lower Limit =			54.530027	=D8-D10*D14		
16	Upper Limit =			57.469973	=D8+D10*D14		
17							
18	For ii) 92% C.I. for Population mean						
19	Here, C.I.(1- α) =			0.92			
20	α =			0.08	=1-D19		
21	Z_{α} =			1.75068607	=NORM.S.INV(1-D20/2)		
22	Here,lower Limit =			54.6869854	=D8-D10*D21		
23	Upper Limit =			57.3130146	=D8+D10*D21		
24							
25	For iii) For probably lie, we should take						
26	Z_{α} =			3			
27	Here,lower Limit =			53.75	=D8-D10*C26		
28	Upper Limit =			58.25	=D8+D10*C26		
29							
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